

ZF Multipliers

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1 Problem Statement: Spatial Lifting of a Hard IQC

We denote the extended L_2 space by $L_{2,e}$ and let $\mathcal{F}$ denote the set of hard factorizations. Suppose that $\Delta : L_{2,e} \rightarrow L_{2,e}$ satisfies the hard IQC associated with a fixed $(\Psi, V) \in \mathcal{F}$,

$$\left\langle p_T \Psi \begin{bmatrix} z \\ \Delta z \end{bmatrix}, p_T V \Psi \begin{bmatrix} z \\ \Delta z \end{bmatrix} \right\rangle_{L_2} \geq 0, \quad (1)$$

for all $T > 0$ and $z \in L_{2,e}$, where $\Psi : L_{2,e}^n \rightarrow L_{2,e}^{\tilde{n}}$ and $V = V^\top \in \mathbb{R}^{\tilde{n} \times \tilde{n}}$.

We extend the signals over the spatial domain $[a, b]$ and define the corresponding signal space as

$$L_{\mathbf{H},e} := \left\{ u : \int_0^T \int_a^b |u(t, s)|^2 ds dt < \infty \text{ for all } T > 0 \right\}. \quad (2)$$

The lifted uncertainty $\hat{\Delta}$ is defined pointwise in the spatial variable by $(\hat{\Delta}u)(\cdot, s) := \Delta(u(\cdot, s))$ for almost every $s \in [a, b]$. Similarly, define the lifted hard factorization $(\hat{\Psi}, \hat{V})$ by $(\hat{\Psi}x)(\cdot, s) := \Psi(x(\cdot, s))$ and $(\hat{V}y)(t, s) := Vy(t, s)$, where $\hat{\Psi} : L_{\mathbf{H},e}^n \rightarrow L_{\mathbf{H},e}^{\tilde{n}}$ and $\hat{V} : L_2^{\tilde{n}}[a, b] \rightarrow L_2^{\tilde{n}}[a, b]$.

The problem is to show that $\hat{\Delta} : L_{\mathbf{H},e} \rightarrow L_{\mathbf{H},e}$ is well-defined and that the temporal hard IQC (1) is preserved under this spatial lifting. Specifically, we seek to show that

$$\left\langle p_T \hat{\Psi} \begin{bmatrix} u \\ \hat{\Delta}u \end{bmatrix}, p_T \hat{V} \hat{\Psi} \begin{bmatrix} u \\ \hat{\Delta}u \end{bmatrix} \right\rangle_{L_{\mathbf{H}}} \geq 0 \quad (3)$$

for all $T > 0$ and $u \in L_{\mathbf{H},e}$.

2 Positive Negative multipliers yield hard factorizations [1]

The lifting of finite dimensional multipliers can be nicely combined with the hard factorization theory developed by Seiler. This hard factorization satisfies the assumptions of the duality theory thus can be directly applied. The main result of [1] is the establishment of

1. (Ψ, M) hard factorizations.

2. $P = P^\top$ is a solution to the KYP lemma. (no enforcement of $P > 0$)

for the class of postive-negative multipliers. Before stating the theorem we need to define J -spectral factors.

Definition 1. (Ψ, M) is called a $J_{n,m}$ -spectral factor of $\Pi = \Pi^\sim \in \mathbb{RL}_\infty^{(n+m) \times (n+m)}$ if $\Pi = \Psi^\sim M \Psi$, $M = \begin{bmatrix} I_n & 0 \\ 0 & -I_m \end{bmatrix}$, and $\Psi, \Psi^{-1} \in \mathbb{RH}_\infty^{(n+m) \times (n+m)}$

Using J -spectral factorizations, Seiler [1] derives the following theorem.

Theorem 2. Let $\Pi = \Pi^\sim$ and partition as $\begin{bmatrix} \Pi_{11} & \Pi_{21} \\ \Pi_{21}^\sim & \Pi_{22} \end{bmatrix}$ where $\Pi_{11} \in \mathbb{RL}_\infty^{n \times n}$ and $\Pi_{22}^{m \times m}$. If $\Pi_{11}(j\omega) > 0$ and $\Pi_{22}(j\omega) < 0$, i.e. Π is strictly positive-negative(PN), $\forall \mathbb{R} \cup \{\infty\}$, then

1. Π has a $J_{n,m}$ -spectral factorization (Ψ, M) .
2. The $J_{n,m}$ -spectral factorization (Ψ, M) is a hard factorization of Π .
3. For $G \in \mathbb{RH}_\infty^{n \times m}$, let $(\hat{A}, \hat{B}, \hat{C}, \hat{D})$ denote the state-space realization of $\Psi \begin{bmatrix} G \\ I \end{bmatrix}$. All solutions $P = P^\top$ to the KYP LMI satisfy $P \geq 0$.

References

- [1] Peter Seiler. Stability analysis with dissipation inequalities and integral quadratic constraints. *IEEE Transactions on Automatic Control*, 60(6):1704–1709, 2015.